6.7 Region-based CNNs (R-CNNs)

- besides single shot multibox detection described in Section 6.6, *region-based CNNs* or regions with CNN features (*R-CNNs*) are also among many pioneering approaches of applying deep learning to object detection
- in this section, we will introduce the R-CNN and its series of improvements: the fast R-CNN, the faster R-CNN, and the mask R-CNN
- the *R-CNN* first extracts many (e.g., 2000) *region proposals* from the input image (e.g., anchor boxes can also be considered as region proposals), labeling their classes and bounding boxes (e.g., offsets)
- then, a CNN is used to perform forward propagation on each region proposal to extract its features
- next, features of each region proposal are used for predicting the class and bounding box of this region proposal

6.7 Region-based CNNs (R-CNNs)

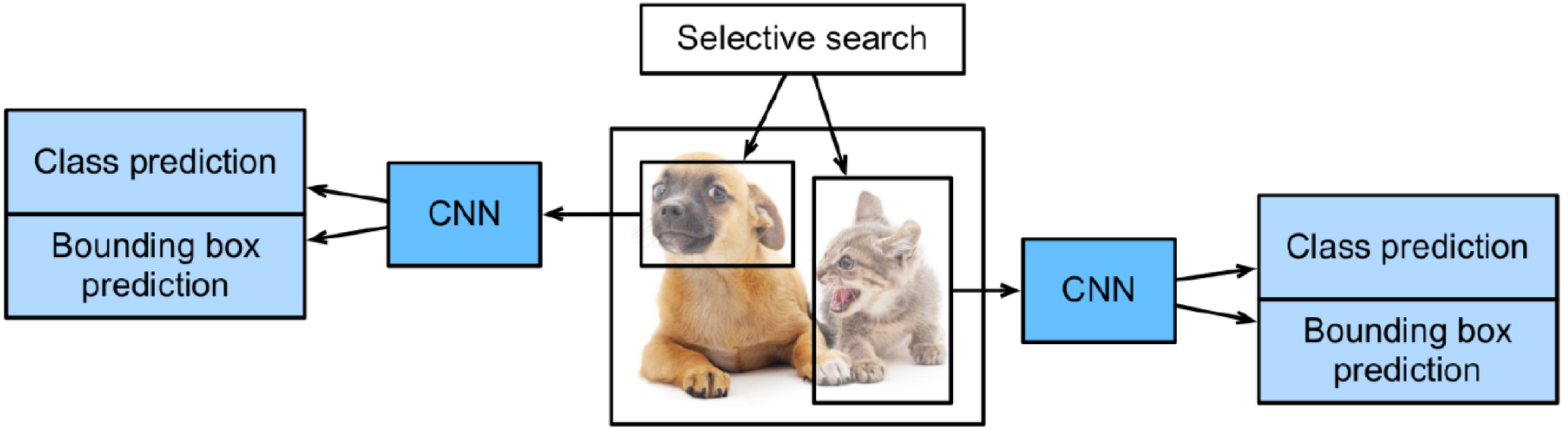


Figure 5: The R-CNN model.

6.7 Region-based CNNs (R-CNNs)

- Figure 5 shows the R-CNN model; more concretely, the R-CNN consists of the following four steps:
  1. Perform *selective search* to extract multiple high-quality region proposals on the input image. These proposed regions are usually selected at multiple scales, with different shapes and sizes. Each region proposal will be labeled with a class and a ground-truth bounding box.
  2. Choose a pretrained CNN, and truncate it before the output layer. Resize each region proposal to the input size required by the network, and output the extracted features for the region proposal through forward propagation.
  3. Take the extracted features and labeled class of each region proposal as an example. Train multiple support vector machines to classify objects, where each support vector machine individually determines whether the example contains a specific class.
  4. Take the extracted features and labeled bounding box of each region proposal as an example. Train a linear regression model to predict the ground-truth bounding box.

6.7 Region-based CNNs (R-CNNs)

- although the R-CNN model uses pretrained CNNs to effectively extract image features, it is slow
- imagine that we select thousands of region proposals from a single input image: this requires thousands of CNN forward propagations to perform object detection
- this massive computing load makes it infeasible to widely use R-CNNs in real-world applications
- the main performance bottleneck of an R-CNN lies in the independent CNN forward propagation for each region proposal, without sharing computation
- since these regions usually have overlaps, independent feature extractions lead to much repeated computation
- one of the major improvements of the *fast R-CNN* from the R-CNN is that the CNN forward propagation is only performed on the entire image

6.7 Region-based CNNs (R-CNNs)

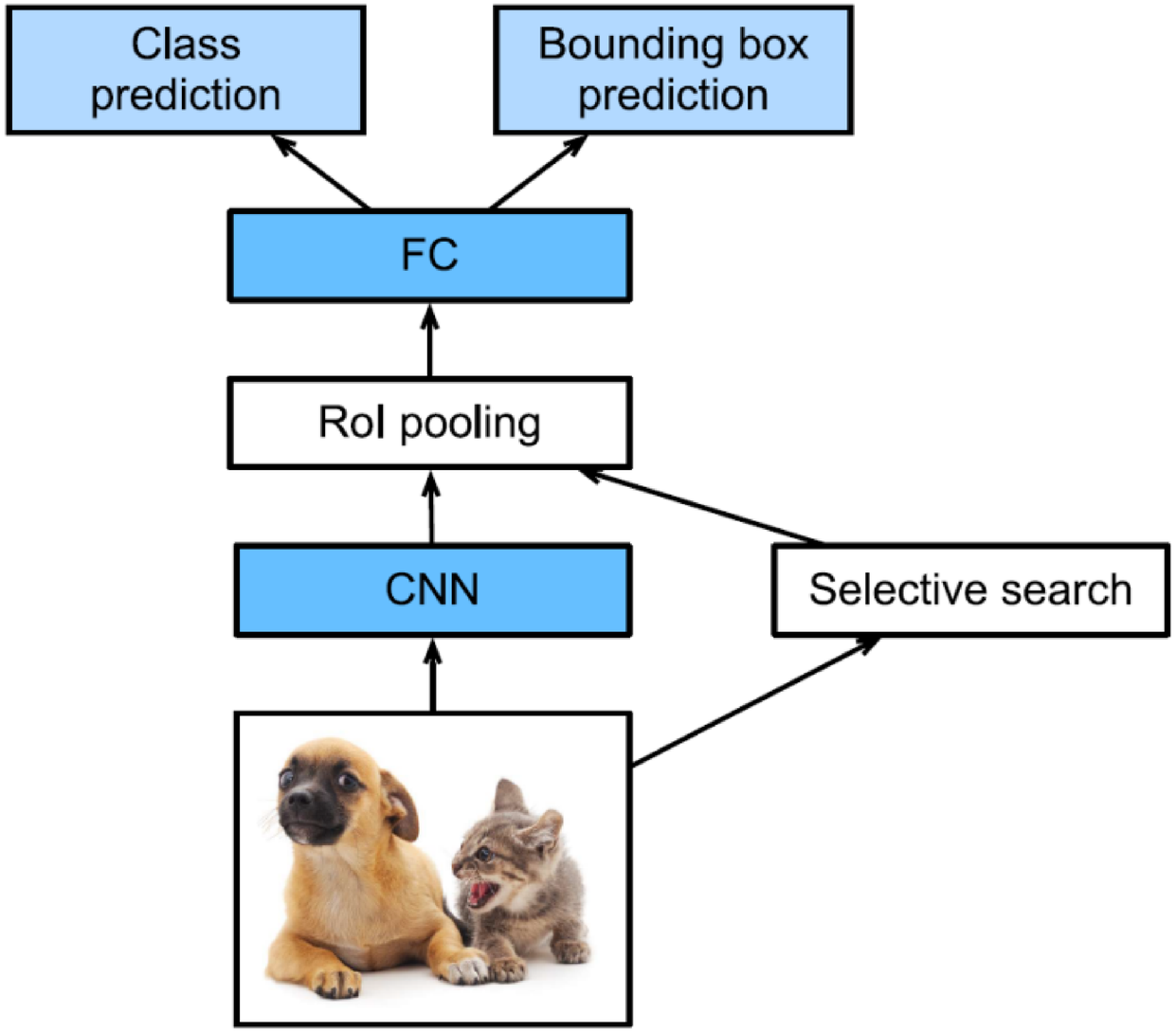


Figure 6: The fast R-CNN model.

6.7 Region-based CNNs (R-CNNs)

- Figure 6 describes the fast R-CNN model; its major computations are as follows:
  1. Compared with the R-CNN, in the fast R-CNN, the input of the CNN for feature extraction is the entire image, rather than individual region proposals. Moreover, this CNN is trainable. Given an input image, let the shape of the CNN output be $1 \times c \times h_1 \times w_1$.
  2. Suppose that selective search generates n region proposals. These region proposals (of different shapes) mark regions of interest (of different shapes) on the CNN output. Then, these regions of interest further extract features of the same shape (say height h_2 and width w_2 are specified) in order to be easily concatenated. To achieve this, the fast R-CNN introduces the *region of interest (RoI) pooling* layer: the CNN output and region proposals are input into this layer, outputting concatenated features of shape $n \times c \times h_2 \times w_2$ that are further extracted for all the region proposals.
  3. Using a fully-connected layer, transform the concatenated features into an output of shape $n \times d$, where d depends on the model design.
  4. Predict the class and bounding box for each of the n region proposals. More concretely, in class and bounding box prediction, transform the fully-connected layer output into an output of shape $n \times q$ (q is the number of classes) and an output of shape $n \times 4$, respectively. The class prediction uses softmax regression.

6.7 Region-based CNNs (R-CNNs)

- the region of interest pooling layer proposed in the fast R-CNN is different from the pooling layer introduced in Section 3.5
- in the pooling layer, we indirectly control the output shape by specifying sizes of the pooling window, padding, and stride
- in contrast, we can directly specify the output shape in the region of interest pooling layer
- for example, let us specify the output height and width for each region as h_2 and w_2, respectively
- for any region of interest window of shape $h \times w$, this window is divided into a $h_2 \times w_2$ grid of subwindows, where the shape of each subwindow is approximately $(h/h_2) \times (w/w_2)$

6.7 Region-based CNNs (R-CNNs)

- in practice, the height and width of any subwindow shall be rounded up, and the largest element shall be used as the output of the subwindow
- therefore, the region of interest pooling layer can extract features of the same shape, even when regions of interest have different shapes
- as an illustrative example, in Figure 7, the upper-left 3×3 region of interest is selected on a 4×4 input
- for this region of interest, we use a 2×2 region of interest pooling layer to obtain a 2×2 output
- note that each of the four divided subwindows contains elements 0, 1, 4, and 5 (5 is the maximum); 2 and 6 (6 is the maximum); 8 and 9 (9 is the maximum); and 10

6.7 Region-based CNNs (R-CNNs)

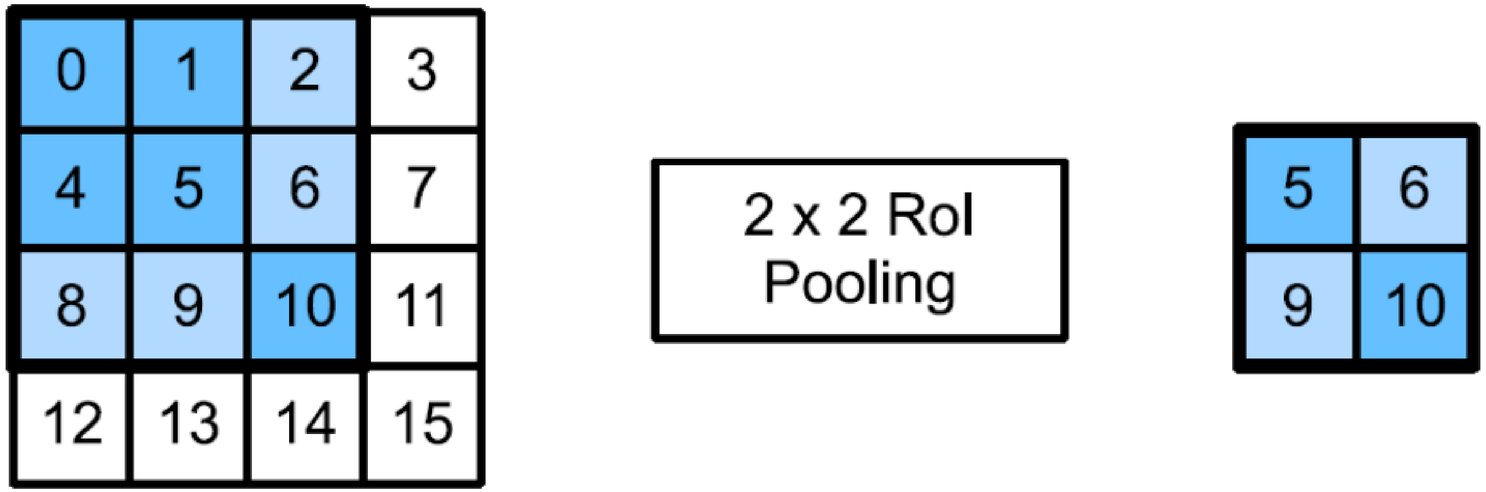


Figure 7: A 2×2 region of interest pooling layer.

6.7 Region-based CNNs (R-CNNs)

- to be more accurate in object detection, the fast R-CNN model usually has to generate a lot of region proposals in selective search
- to reduce region proposals without loss of accuracy, the *faster R-CNN* proposes to replace selective search with a *region proposal network*

6.7 Region-based CNNs (R-CNNs)

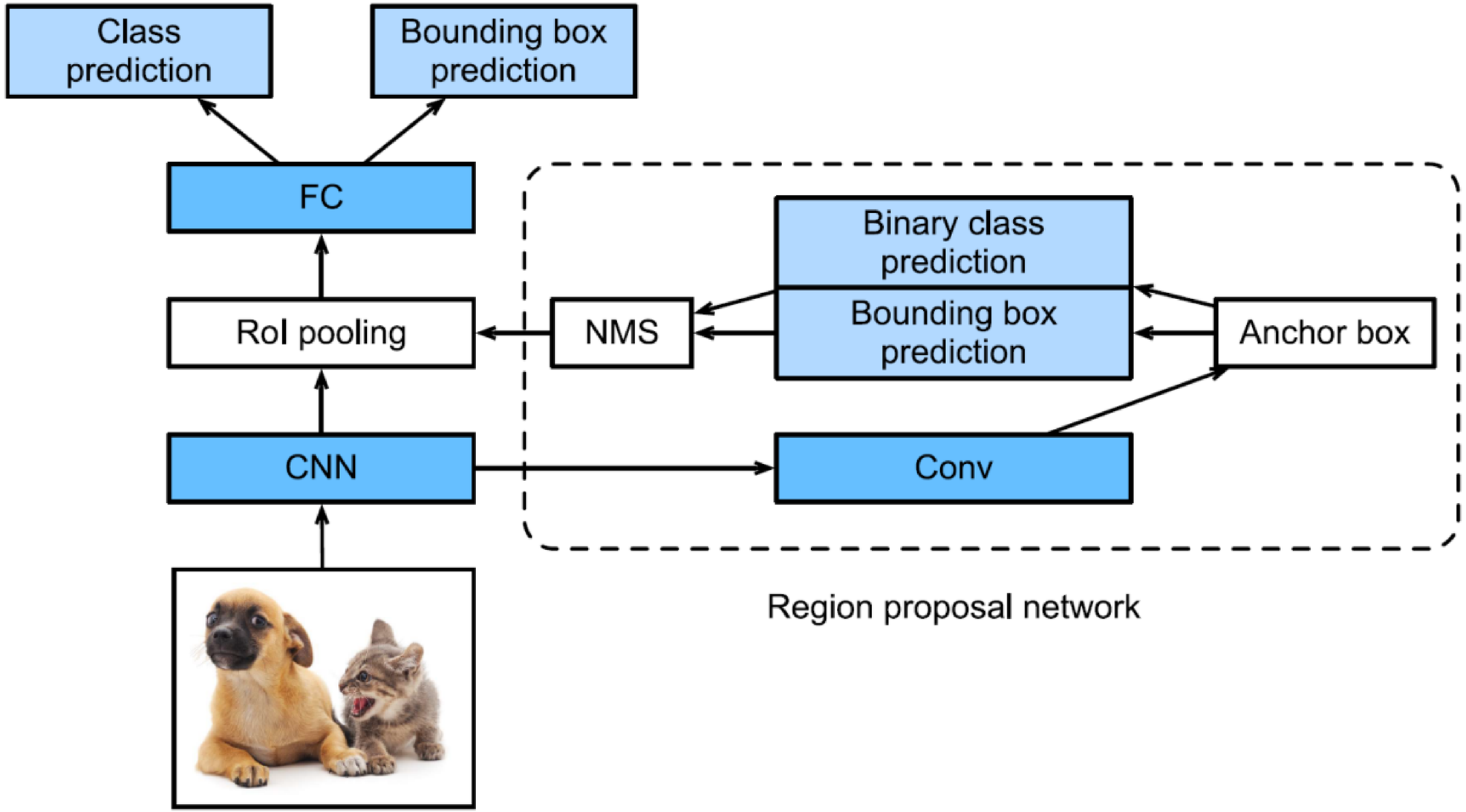


Figure 8: The faster R-CNN model.

6.7 Region-based CNNs (R-CNNs)

- Figure 8 shows the faster R-CNN model
- compared with the fast R-CNN, the faster R-CNN only changes the region proposal method from selective search to a region proposal network
- the rest of the model remains unchanged; the region proposal network works in the following steps:
  1. Use a 3×3 convolutional layer with padding of 1 to transform the CNN output to a new output with c channels. In this way, each unit along the spatial dimensions of the CNN-extracted feature maps gets a new feature vector of length c.
  2. Centered on each pixel of the feature maps, generate multiple anchor boxes of different scales and aspect ratios and label them.
  3. Using the length-c feature vector at the center of each anchor box, predict the binary class (background or objects) and bounding box for this anchor box.
  4. Consider those predicted bounding boxes whose predicted classes are objects. Remove overlapped results using non-maximum suppression. The remaining predicted bounding boxes for objects are the region proposals required by the region of interest pooling layer.

6.7 Region-based CNNs (R-CNNs)

- it is worth noting that, as part of the faster R-CNN model, the region proposal network is jointly trained with the rest of the model
- in other words, the objective function of the faster R-CNN includes not only the class and bounding box prediction in object detection, but also the binary class and bounding box prediction of anchor boxes in the region proposal network
- as a result of the end-to-end training, the region proposal network learns how to generate high-quality region proposals, so as to stay accurate in object detection, with a reduced number of region proposals that are learned from data
- in the training dataset, if pixel-level positions of objects are also labeled on images, the *mask R-CNN* can effectively leverage such detailed labels to further improve the accuracy of object detection

6.7 Region-based CNNs (R-CNNs)

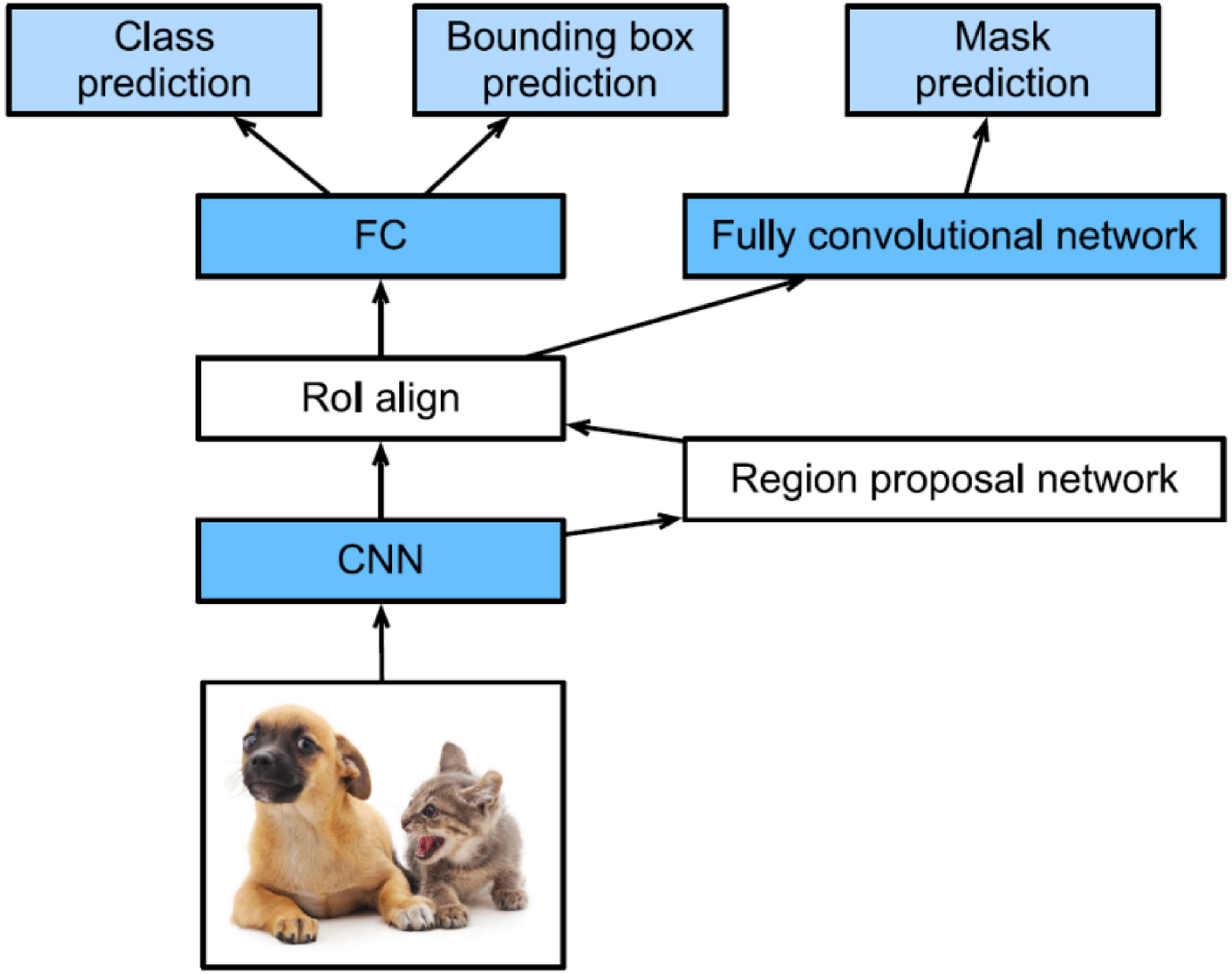


Figure 9: The mask R-CNN model.

6.7 Region-based CNNs (R-CNNs)

- as shown in Figure 9, the mask R-CNN is modified based on the faster R-CNN
- specifically, the mask R-CNN replaces the region of interest pooling layer with the *region of interest* (*RoI*) *alignment* layer
- this region of interest alignment layer uses bilinear interpolation to preserve the spatial information on the feature maps, which is more suitable for pixel-level prediction
- the output of this layer contains feature maps of the same shape, for all the regions of interest
- they are used to predict not only the class and bounding box for each region of interest, but also the pixel-level position of the object, through an additional fully convolutional network
- more details on using a fully convolutional network to predict pixel-level semantics of an image will be provided in the next chapter